

Supplementary Material for “3D Embodied Agent-Mediated Affective Art Interaction for Emotion Externalization and Regulation”

Xiang Li, Lei Meng, Shiyu Zhu, Zongyi Yang, Yachong Zhang, Fangyuan Wei, Chunbo Li, and Xiangxu Meng

S1. DETAILED GAME-STATE ENCODING

Let C_t , O_t , and R_t denote the canvas state, recent operation sequence, and reconstruction trajectory, respectively. The interpretable game state is

$$w_t = \phi_g(C_t, O_t, R_t) = [q_t^c, q_t^o, q_t^r], \quad q_t^r = q_t^c - q_0^c. \quad (S1)$$

where $\phi_g(\cdot)$ maps composition and interaction behavior into interpretable contextual cues. The canvas descriptor is

$$q_t^c = [n_t, \chi_t, \eta_t, \alpha_t, \kappa_t, \mu_t, \nu_t], \quad (S2)$$

where the entries denote shape distribution, dominant color, density, contour sharpness, blank-space ratio, spatial balance, and repetition or rhythm. The interaction descriptor is

$$q_t^o = [e_t, \tau_t, z_t, \xi_t], \quad (S3)$$

where the entries denote recent edits, idle duration, undo or repeated-modification frequency, and accepted system suggestions. The descriptor q_t^r records the change from the initial composition.

These cues are not assigned fixed psychological meanings. They are combined with direct speech/text and confidence-weighted video evidence and are used only as contextual interaction evidence.

S2. RESPONSE PLANNING, RELIABILITY, AND PROVENANCE

A. Illustrative End-to-End Walkthrough

The following walkthrough demonstrates the system workflow and is not a verbatim participant transcript. The avatar begins, “Please use the available colors and shapes to show how you feel right now; there is no right or wrong composition.” The user replies, “I feel tense, and everything seems crowded today,” and creates the dense arrangement of overlapping red and orange triangles, strong diagonal lines, high contrast, and limited blank space shown in Fig. 6(a) of the main manuscript. The state summary records the explicit report of tension, moderate-confidence negative-affect video evidence, and a dense, sharp, high-contrast canvas, while treating the video and visual cues as contextual rather than diagnostic. The planner selects validation and grounding. In a slower voice, the avatar says, “It sounds as though things feel crowded right now. We can slow down. Would you like to open one small area and soften one shape?” while slightly nodding and adopting an open, relaxed posture. The user separates the elements, replaces several sharp triangles with circles and

rounded rectangles, lowers the contrast, and increases blank space, producing Fig. 6(b). The avatar closes by neutrally reflecting the visible changes without assigning emotional meaning to the artwork.

The strategy vocabulary is predefined rather than learned from user-study outcomes. Validation, reassurance, reflection, and encouragement were adapted from emotional-support dialogue taxonomies [1]–[4]; grounding is a low-risk interaction heuristic, and creative guidance is a task-specific heuristic. The prompt-constrained planner selects from the structured state summary and evidence hierarchy, with game guidance optional. This is a non-clinical support policy, not a learned therapeutic policy.

B. Module Reliability and Fallbacks

Module outputs carry availability and confidence flags rather than forming simultaneous hard prerequisites. The planner uses any valid subset, never fabricates absent evidence, and never converts fallback behavior into a deterministic emotion label.

C. Component Provenance

Integrated components are React, Capacitor, Three.js, and Fay for the interface; Hunyuan3D for mesh generation [5]; Grounded SAM 2 for masks [6], [7]; the S4D checkpoint and backbone for video initialization [8]; the UniRig base pipeline for rigging [9]; Mixamo motions [10]; and Fun-ASR, Qwen3-VL-Flash/DashScope, and Fun-CosyVoice for speech, inference, and synthesis [11], [12]. Our contributions are (i) the composition game, game-state encoding, and optional guidance; (ii) semantic facial-region tokenization, semantic-global interaction, and semantic-conditioned MoE routing; (iii) structure-aware binding encoding and multi-skeleton consistency fusion; and (iv) the multimodal evidence hierarchy, support-strategy planning, and multi-channel response coordination. We do not claim the upstream models, services, frameworks, or motion assets as original contributions.

S3. AVATAR RIGGING FAILURE BOUNDARY

The prototype admitted only successfully generated and rigged humanoids and has no runtime recovery or default-avatar fallback. Extreme proportions, non-humanoid anatomy, missing or disconnected parts, irregular or nonmanifold topology, self-intersections, and loose accessories can cause

TABLE S1
INTERPRETABLE GAME-STATE FEATURES EXTRACTED FROM THE AFFECTIVE-ART COMPOSITION GAME.

Feature	Definition	Extraction method	Interaction intuition
Shape distribution	Number, type, size, and spatial distribution of primitives	Primitive metadata and canvas coordinates	Expressive complexity and visual focus
Dominant color	Hue, saturation, and brightness distribution	Area-weighted color histogram	Affective tone and intensity
Visual density	Occupied element area divided by canvas area	Geometry mask or summed primitive area	Crowdedness and expressive pressure
Contour sharpness	Sharpness implied by primitive type and corner angles	Primitive labels and polygon angles	Visual tension or softness
Blank-space ratio	Empty canvas area divided by total canvas area	Complement of the occupancy mask	Openness and breathing room
Spatial balance	Distance between visual centroid and canvas center	Area-weighted centroid	Stability or imbalance
Repetition/rhythm	Repeated primitive, color, or spatial patterns	Histograms and repeated-operation counts	Regularity and persistence
Editing dynamics	Edit types, idle duration, undo frequency, and accepted guidance	Operation-log sequence	Hesitation, exploration, and reconstruction

TABLE S2
MODULE-SPECIFIC RELIABILITY AND FALLBACK BEHAVIOR.

Module	Unavailable or low-confidence state	Fallback	Safety boundary
Facial video	Missing or low-confidence evidence	Exclude video and continue with valid speech/text and game state	Do not infer missing affect
ASR	Recognition failure	Ask for repetition or switch to typed input	Do not guess the utterance
Game state	Canvas or operation state unavailable	Continue dialogue and disable canvas-specific guidance	No artwork interpretation
Qwen3-VL-Flash	Timeout or service failure	Display a restricted neutral support template and allow retry	No clinical advice or certain label
TTS	Synthesis failure	Keep the text response visible and skip voice output	Preserve response content
Animation	Motion-generation or retargeting failure	Skip the gesture channel and retain text/available voice	Avoid unstable motion

joint mislocalization, weight leakage, or motion penetration. Broader style benchmarks, pre-rig mesh validation, failure detection, and reject, regenerate, or default-avatar recovery are required.

S4. USER-STUDY PROTOCOL AND INSTRUMENTS

A. Counterbalancing and Cross-Day Controls

The four cyclic Latin-square sequences were C1–C2–C3–C4, C2–C3–C4–C1, C3–C4–C1–C2, and C4–C1–C2–C3. Six participants followed each sequence, and the remaining participant was assigned to one sequence, producing a 6/6/6/7

allocation. Each participant’s sessions were scheduled at approximately the same time of day. One condition per day reduced immediate carry-over and fatigue; the Latin square distributed condition order and adaptation; and daily pre/post measures supplied session-specific baselines. These controls cannot eliminate daily mood variation, time-of-day differences, or adaptation.

B. Post-Session Questionnaire

The administered questionnaire was in Chinese, used a 1–5 Likert scale, and contained the following 11 closed-form items. English translations are provided for reporting transparency.

TABLE S3
COMPLETE POST-SESSION QUESTIONNAIRE ITEMS.

Dimension	Original Chinese item	English translation
Empathy	智能体准确捕捉到我通过视频表达的情绪	The agent accurately captured the emotion I expressed through video.
Empathy	智能体的建议和回复自然, 不像预设脚本	The agent’s suggestions and responses were natural rather than scripted.
Empathy	智能体理解我产生这种情绪的原因	The agent understood why I felt this emotion.
Empathy	回复比普通聊天机器人更有温度与深度	The responses felt warmer and more in-depth than those of an ordinary chatbot.
Social Presence and Affinity	感觉在与有生命的实体交流, 而非机器	I felt that I was interacting with a living entity rather than a machine.
Social Presence and Affinity	3D 形象动作表情有效传达关注	The 3D avatar’s movements and expressions effectively communicated attentiveness.
Social Presence and Affinity	3D 形象让人放松, 愿意倾诉内心	The 3D avatar made me feel relaxed and willing to share my inner feelings.
Task Experience	创作时沉浸投入, 忘记时间流逝	I was immersed in the creative activity and lost track of time.
Task Experience	艺术创作轻松, 无“必须画好”压力	The art-making felt relaxed, without pressure that I had to make it good.
Task Experience	能用色彩图形将情绪具象化表达	I could concretize and express my emotions through colors and shapes.
Task Experience	能够有效缓解紧张、焦虑等负面情绪, 带来平静与放松感。	The activity effectively relieved tension or anxiety and brought a sense of calm and relaxation.

The administered instruments contained no open-ended comment field, and no qualitative interview transcript was available. Consequently, no verbatim participant comments can

be reported. Future studies should add open-ended prompts and semistructured interviews.

C. Privacy Boundary

Speech/text, facial video, canvas/action logs, and questionnaires were treated as sensitive affective data. Participants consented to these modalities, and anonymized records were used for the stated analyses. Facial and canvas signals were contextual, non-diagnostic evidence. The study does not establish on-device processing or a formal privacy guarantee. Future deployments should specify access control, encryption in transit and at rest, retention and deletion schedules, withdrawal procedures, and release only de-identified derived data.

REFERENCES

- [1] S. Liu, C. Zheng, O. Demasi, S. Sabour, Y. Li, Z. Yu, Y. Jiang, and M. Huang, "Towards emotional support dialog systems," in *ACL-IJCNLP*, 2021, pp. 3469–3483.
- [2] Q. Tu, Y. Li, J. Cui, B. Wang, J.-R. Wen, and R. Yan, "Misc: A mixed strategy-aware model integrating comet for emotional support conversation," in *ACL*, 2022, pp. 308–319.
- [3] J. Zhou, Z. Chen, B. Wang, and M. Huang, "Facilitating multi-turn emotional support conversation with positive emotion elicitation: A reinforcement learning approach," in *ACL*, 2023, pp. 1714–1729.
- [4] Y. Chen, X. Xing, J. Lin, H. Zheng, Z. Wang, Q. Liu, and X. Xu, "Soulchat: Improving llms' empathy, listening, and comfort abilities through fine-tuning with multi-turn empathy conversations," in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023, pp. 1170–1183.
- [5] Z. Zhao, Z. Lai, Q. Lin, Y. Zhao, H. Liu, S. Yang, Y. Feng, M. Yang, S. Zhang, X. Yang *et al.*, "Hunyuan3d 2.0: Scaling diffusion models for high resolution textured 3d assets generation," *arXiv preprint arXiv:2501.12202*, 2025.
- [6] T. Ren, S. Liu, A. Zeng, J. Lin, K. Li, H. Cao, J. Chen, X. Huang, Y. Chen, F. Yan *et al.*, "Grounded sam: Assembling open-world models for diverse visual tasks," *arXiv preprint arXiv:2401.14159*, 2024.
- [7] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson *et al.*, "Sam 2: Segment anything in images and videos," 2024.
- [8] Y. Chen, J. Li, Y. Zhang, Z. Hu, S. Shan, M. Wang, and R. Hong, "Static for dynamic: Towards a deeper understanding of dynamic facial expressions using static expression data," *IEEE Transactions on Affective Computing*, 2025.
- [9] J.-P. Zhang, C.-F. Pu *et al.*, "One model to rig them all: Diverse skeleton rigging with UniRig," *ACM Transactions on Graphics*, vol. 44, no. 4, pp. 1–18, 2025.
- [10] S. Blackman, "Rigging with mixamo," in *Unity for Absolute Beginners*. Springer, 2014, pp. 565–573.
- [11] Z. Gao, Z. Li, J. Wang, H. Luo, X. Shi, M. Chen, Y. Li, L. Zuo, Z. Du, Z. Xiao *et al.*, "Funasr: A fundamental end-to-end speech recognition toolkit," 2023.
- [12] Z. Du, Q. Chen, S. Zhang, K. Hu, H. Lu, Y. Yang, H. Hu, S. Zheng, Y. Gu, Z. Ma *et al.*, "Cosyvoice: A scalable multilingual zero-shot text-to-speech synthesizer based on supervised semantic tokens," *arXiv preprint arXiv:2407.05407*, 2024.